

VENKATA SRI SAI SURYA MANDAVA

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TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C++, Bash

Frameworks & Architecture: Node.js, Express, FastAPI, Flask, React, Next.js, Spring Boot, REST APIs, WebSockets, Microservices

Cloud, DevOps & Databases: AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Git, Linux, Redis, PostgreSQL, MySQL, MongoDB

AI & Data: OpenAI API, LangChain, LangGraph, AI Agents, RAG, Vector DBs (Pinecone), Scikit-learn, Pandas, NumPy

Certifications: AWS Certified Solutions Architect – Associate, Microsoft Certified – Azure Fundamentals

EDUCATION

The State University of New York at Buffalo, Buffalo, New York

Aug. 2024 – Jan. 2026

Master of Science in Computer Science and Engineering

GPA: 3.76/4.0

Coursework: Algorithms, Machine & Deep Learning, Data Mining, Operating Systems, Databases, Distributed Systems, Security, Computer Vision

Velagapudi Ramakrishna Siddhartha Engineering College, Vijayawada, India

Aug. 2020 – April 2024

Bachelor of Technology in Information Technology

GPA: 8.78/10

EXPERIENCE

Crypt0nest.io LLC, Austin, Texas, United States

June 2025 – Aug. 2025

AI/ML R&D and Product Intern (Team Lead)

- Built an AI-integrated analytics platform with Python, Node.js, and SQL APIs to process real-time market signals for 250+ crypto assets across 3 product teams
- Engineered ML feature pipelines (RSI, MACD, momentum, OBV) using CoinGecko, DeFiLlama, and LunarCrush data, improving prediction accuracy by ~30% with SHAP-based explainability
- Deployed ML inference services on AWS using Docker and CI/CD, achieving ~100ms prediction latency and reducing release cycles by 40%
- Implemented monitoring, health checks, and automated rollbacks for ML inference services, reducing production failures and improving prediction reliability

I&T LABS, Vijayawada, India

May 2023 – July 2024

Software Engineer

- Built a full-stack academic platform using Node.js, PostgreSQL, and REST APIs to manage 5,000+ records, automating workflows and reducing manual work by 20+ hrs/week
- Developed async backend services with OAuth2/JWT authentication, maintaining API latency around ~250ms under peak usage
- Optimized PostgreSQL schema, indexing, queries to support 1,000+ concurrent users, enabling real-time KPI dashboards, analytics insights
- Implemented logging, error handling, service monitoring, improving system stability and supporting data-driven features across the platform

Codegnan IT Solutions Pvt Ltd, Vijayawada, India

Aug. 2022 – April 2023

Software Engineer

- Built a MERN-based social platform with real-time WebSocket chat, supporting 2,000+ active users and enabling scalable messaging services
- Developed 10+ secure REST APIs using Node.js, Express, and OAuth2/JWT for authentication and core platform functionality
- Containerized backend services with Docker and standardized deployment workflows, enabling weekly production releases across 3+ environments
- Refactored backend request pipelines and async processing workflows, improving request throughput and reducing runtime failures during peak usage

PROJECTS

RoomieConnect | Full-Stack Cloud Expense Management System

June 2025 – Sep. 2025

- Built an AI-integrated expense management app with React.js, Node.js, and PostgreSQL, automating shared expense tracking with 98% transaction accuracy
- Integrated the OpenAI API for automated expense categorization and insights, reducing manual entry by ~60%
- Deployed containerized services on AWS EC2 with Docker Compose and S3, supporting 200+ concurrent users with 99%+ uptime
- Implemented Stripe payment APIs and role-based access control, reducing payment failures by 45% and speeding feature rollout by 20%

Crime Analytics System

Jan. 2025 – May 2025

- Built a backend crime analytics system using Python APIs and PostgreSQL, managing 50,000+ records for spatial crime analysis
- Developed automated ETL pipelines and optimized SQL queries, reducing data duplication by 60% and improving query performance by 40%
- Implemented AI-assisted hotspot detection and built real-time Tableau dashboards, identifying crime patterns with ~85% accuracy

Blood Sugar Regulatory System

June 2023 – April 2024

- Built AI-integrated backend services to process real-time CGM and biosensor streams, improving prediction accuracy from 65% to 92%
- Developed scalable data ingestion and validation pipelines, reducing manual monitoring by 50% under live workloads
- Implemented automated CI/CD and testing workflows for AI inference services, reducing deployment effort by 35% and improving reliability